engineering 4 us all

Student Handout

Name: ______________________________ **Date:** _________________ **Class:** _______________

MyDesign Portfolio

Team 1 element B

Element B: Documentation and Analysis of Prior Design Attempts

Prior Design Attempt #1 - <drip tubing

Source

https://school.sprinklerwarehouse.com/design-install/drip-tubing-basics-and-drip-irrigation-system-basics/#:~:text=What%20Is%20Drip%20Tubing%3F,the%20placement%20of%20each%20plant.

Image

https://www.gardeners.com/buy/garden-row-snip-n-drip-soaker-system/8587042.html

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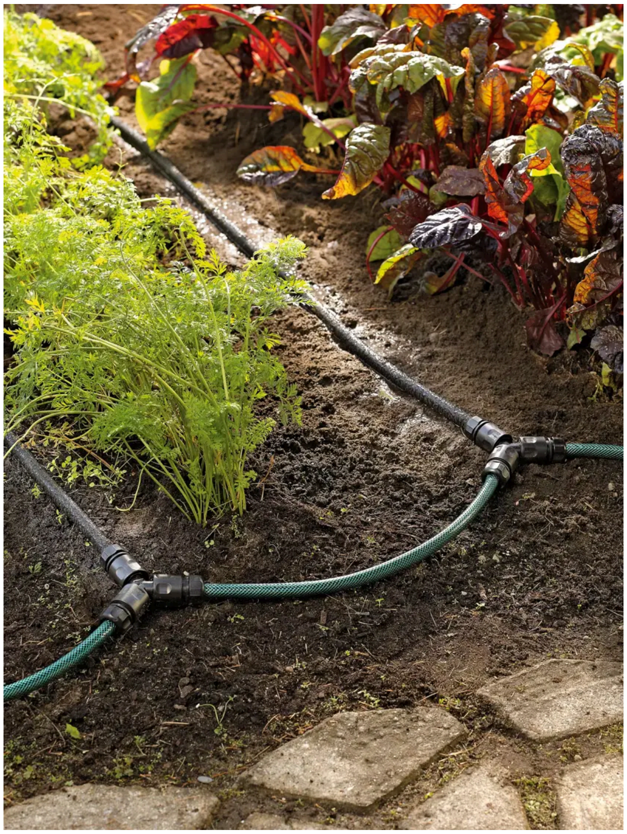

Details

Drip tubing is when water flows through polyurethane tubing and is dispersed into plants by emitters. Drip tubing is a very efficient method of watering plants as it has over 90% irrigation rate. It also reduces disease rates as the tubes sends water directly to the roots of the plants

Analysis

This design was unsuccessful because although drip tubing is good at irrigation plants it is not easy to learn how to use and install and is not the most optimal method for preventing mold.

Prior Design Attempt #2 - <insert name of prior design attempt #2>

Source

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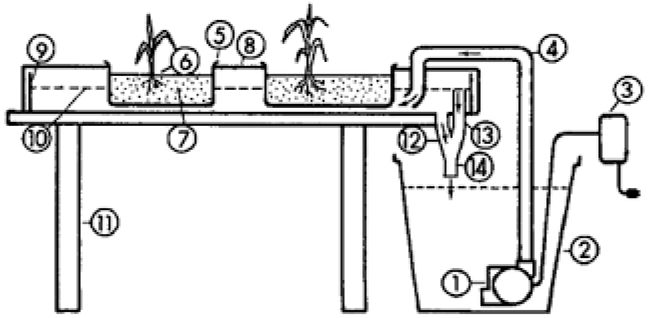
https://ag.umass.edu/greenhouse-floriculture/fact-sheets/subirrigation-systems#:~:text=In%20subirrigation%20systems%2C%20water%20and,grown%20in%20pots%20or%20flats.

Image

https://www.researchgate.net/figure/A-fully-automatic-subirrigation-system-for-greenhouse-and-growth-chamber-use-1_fig6_279925628



Details

Nutrients and water are provided at the base of the plant containers and rises by capillary action through holes in the container to help grow the plants.

Analysis

This method is good at growing plants uniformly and is very good at preventing runoff and wasted water however mold can grow within the containers and it is possible that the device breaks which would allow mold to grow.

Prior Design Attempt #3 - Overhead greenhouse sprinkler

Source

https://www.shinygrow.com/products/grenhouse-sprinkler.html

Image

https://www.shinygrow.com/products/grenhouse-sprinkler.html

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Details

It is an automatic overhead misting system which irrigates plants by sprawling water on top of them. This system is good at increasing the humidity of the greenhouse as well as clearing dust from the air

Analysis

This design was not successful because more humid environments were more likely to promote mold growth and mold could grow inside the machine. The system is also automatic meaning students would not be able to interact with it much and if it broke that could be very problematic, it is also very expensive.

Rubric

Element B: Documentation and Analysis of Prior Design Attempts						
	score = 5	score = 4	score = 3	score = 2	score = 1	score = 0
B1. Documentation of plausible prior attempts to solve the problem and/or related problems is drawn from ___________.	a wide array of clearly-identified and consistently-credible sources	a variety of clearly-identified and consistently-credible sources	several—but not necessarily varied—clearly-identified and generally-credible sources	a limited number of sources, some of which may not be clearly identified and/or credible	only one or two sources that may not be clearly identified and/or credible	missing or minimal (a single source that is not clearly identified and/or credible)
B2. The analysis of past and current attempts to	score = 5	score = 4	score = 3	score = 2	score = 1	score = 0

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solve the problem — including both strengths and shortcomings — __________.	is consistently clear, detailed, and supported by relevant data	is clear, generally detailed, and supported by relevant data	is generally clear and contains some detail and relevant supporting data	is overly general and contains little detail and/or relevant supporting data	is vague and is missing any relevant details and/or relevant supporting data	cannot be inferred from information intended as analysis of past and/or current attempts to solve the problem